Educational Ludo - Interactive Prototype

Assignment 3, part 1: prototype | presented in class on 02/11/2026

Team Sharks

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About the prototype

A Java 17 Swing desktop application that runs completely offline (EducationalLudo.jar, started with run.bat). Screens show simulated data; the dice, piece moves, questions and surprises follow a simplified version of the SRS rules. A small, clearly labelled PROTOTYPE DEMO panel lets testers open a question, a surprise or the results without playing a full game. Public link: not applicable - the prototype is a local desktop application provided with the submission.

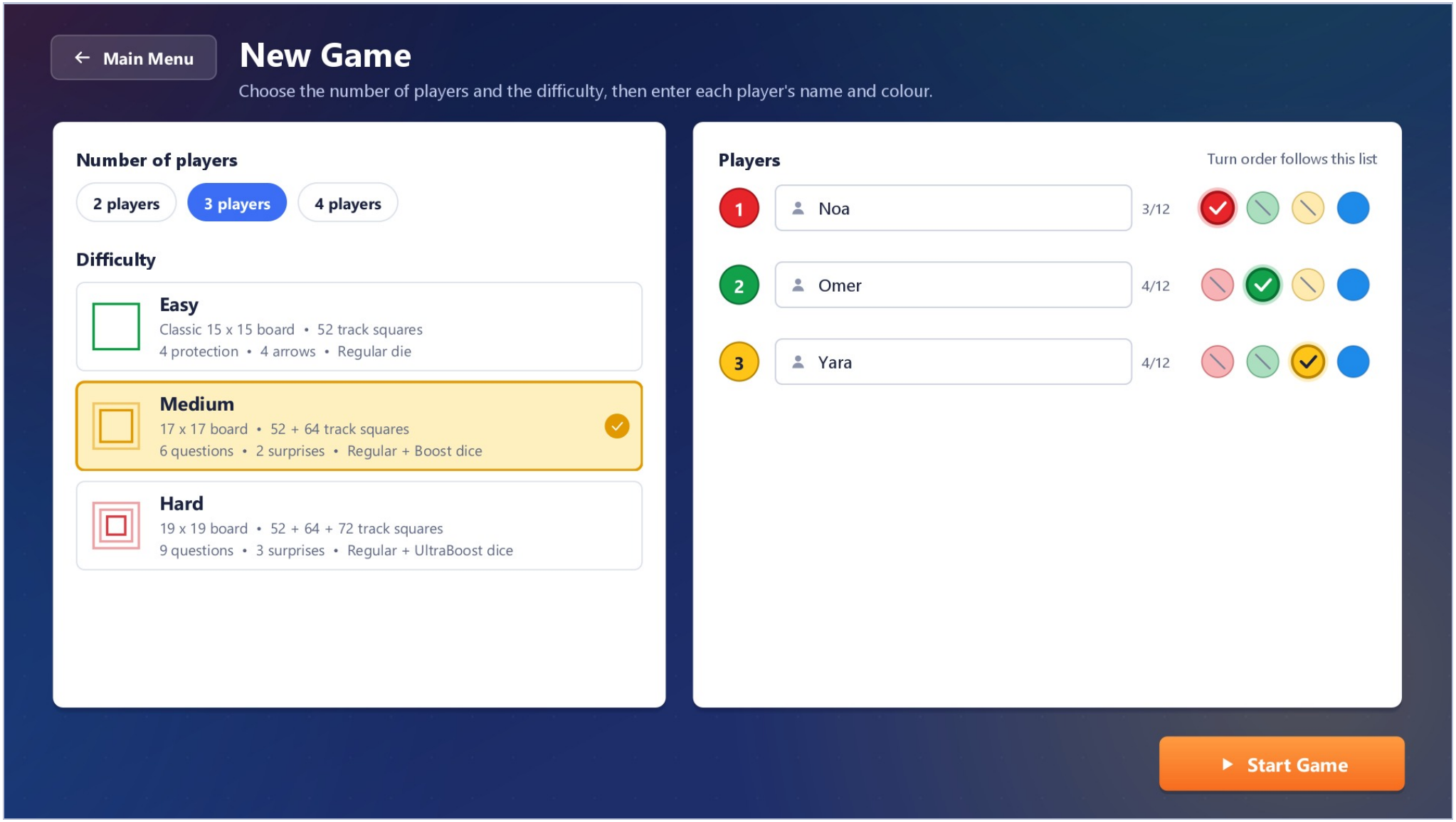
1. Main Menu

The application opens on the main menu: New Game, Game History, Question Management (administrators only, behind a login) and Exit. (FR-01, FR-02)



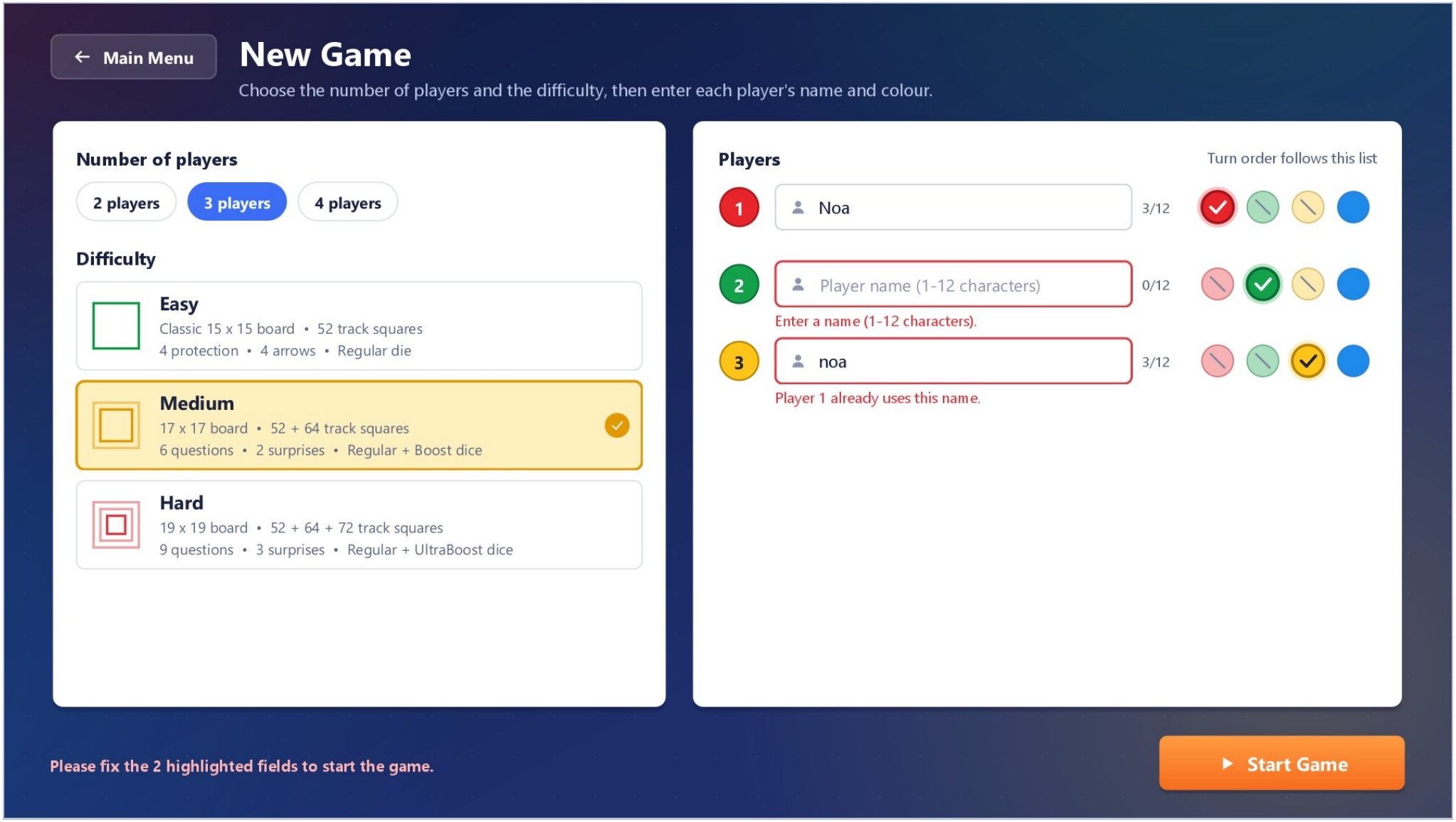
2. Game Setup

Choose 2, 3 or 4 players and the difficulty; each card explains the board, special squares and dice of the level. Every player enters a unique name (1-12 characters) and picks a unique colour; colours taken by another player are disabled. (FR-04 - FR-07)



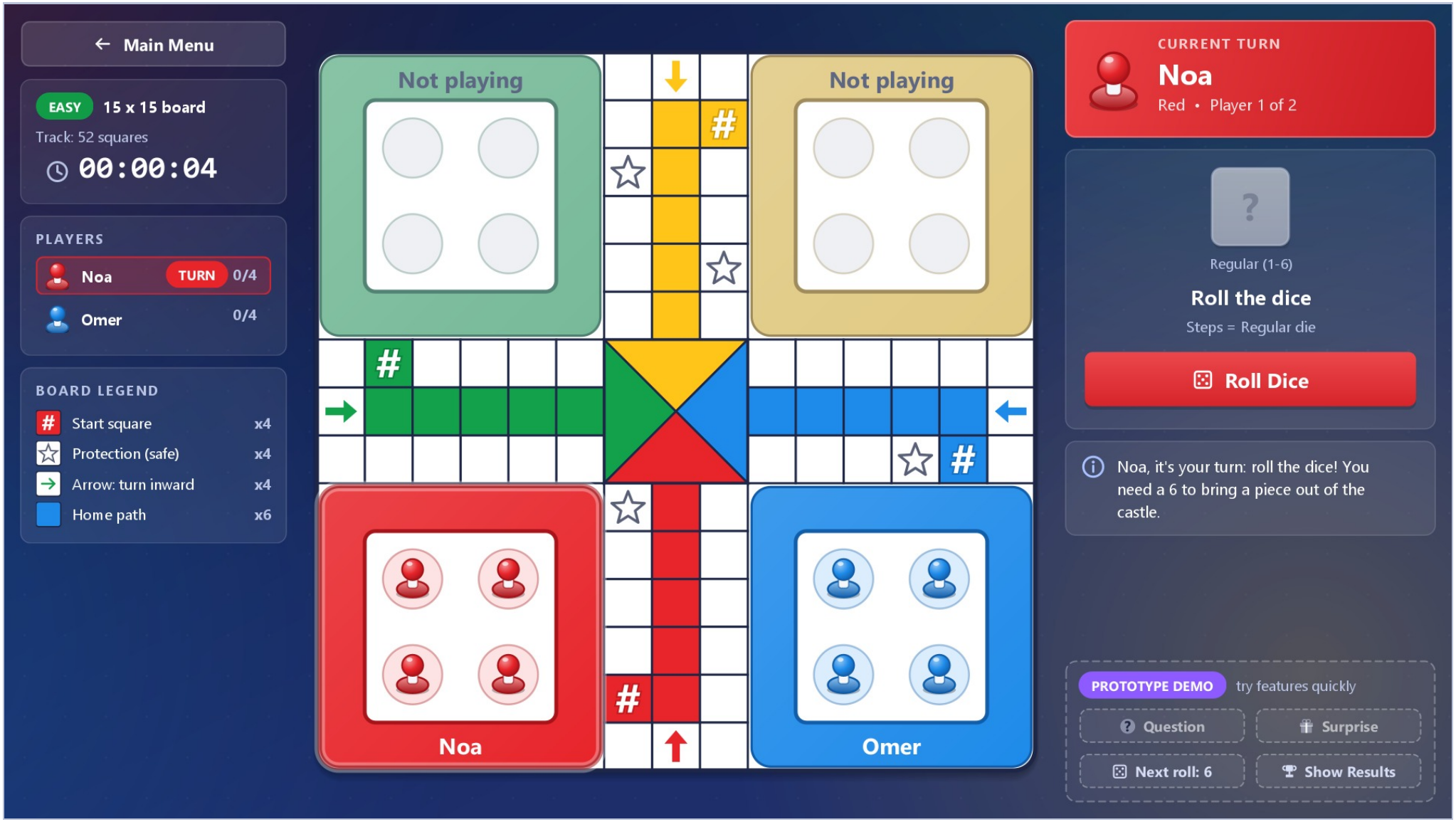
3. Setup Validation

Invalid input is caught before the game starts: the empty name and the duplicate name are highlighted with a message next to the field. (FR-08)



4. Game Board - Easy

Easy board (classic 15 x 15, drawn like the official diagram): castles with four pieces, the 52-square track, coloured home paths, # start squares, arrows and protection stars. Left: level, timer, players and legend. Right: whose turn it is, the dice and Roll Dice. Easy has no question or surprise squares, so those two demo buttons are off. (FR-10 - FR-15, FR-46)



5. Game Board - Hard

Hard board (19 x 19): two extra rings of 64 and 72 squares, start squares in the outer corners, three arrows per colour leading inward ring by ring, question squares coloured by question level (green / yellow / red) and purple surprise squares. (SRS Table 1)



6. Game Board - Medium

Medium board (17 x 17, 52 + 64 squares) during a game: Regular + Boost dice, 8 protection squares, 8 arrows, 6 question squares and 2 surprise squares.



7. Roll Dice

After a short dice animation the result and the movement Regular x (Boost - 1) are shown. The pieces that can move glow; pointing at one previews its path and landing square. (FR-16, FR-17, FR-24)



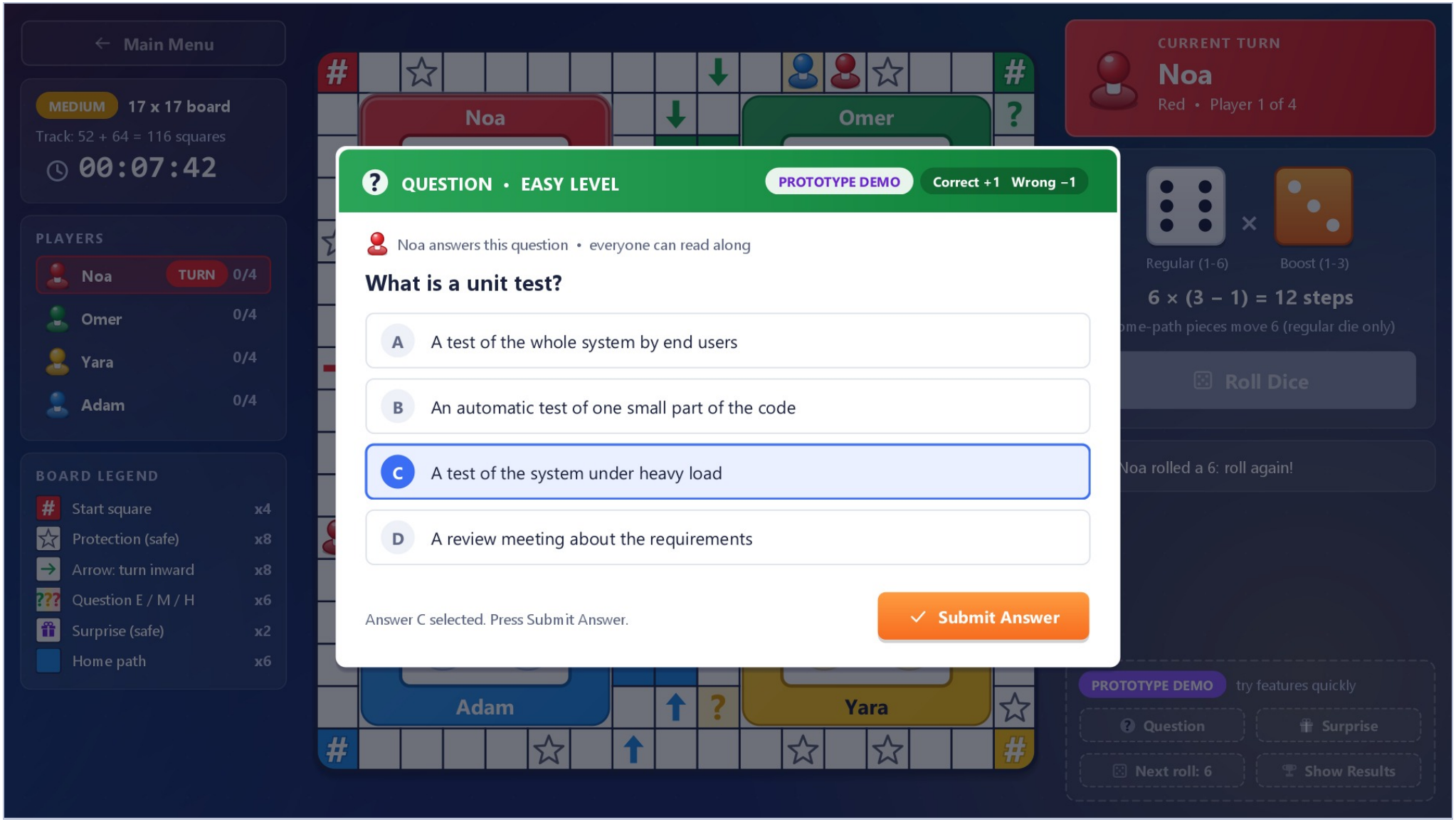
8. Piece Moved

Clicking a glowing piece moves it square by square. A 6 was rolled, so the same player rolls again. (FR-20, FR-21)



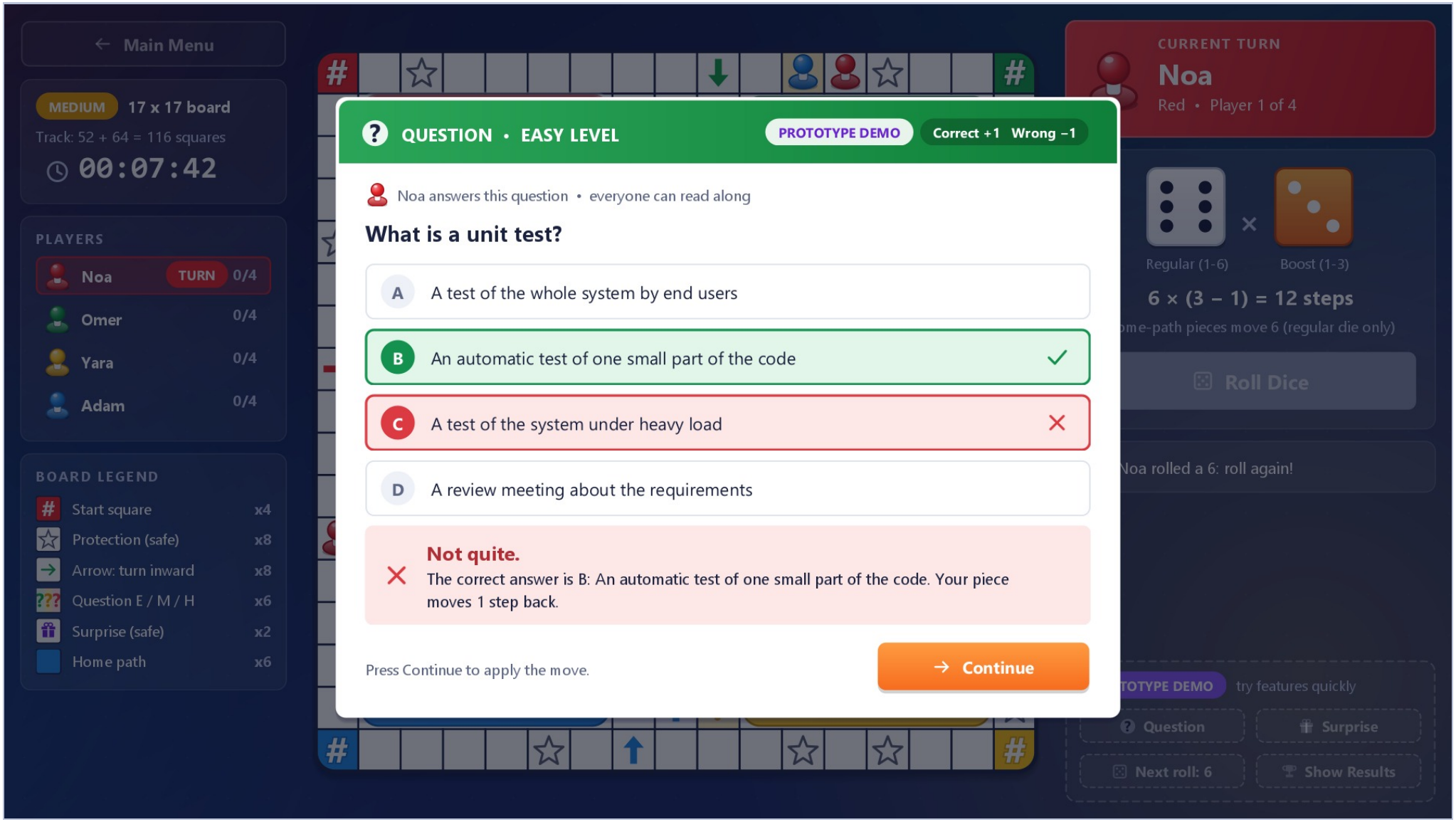
9. Educational Question

Landing on a question square opens a multiple-choice question of that square's level. The active player selects one of four answers and submits it. (FR-34)



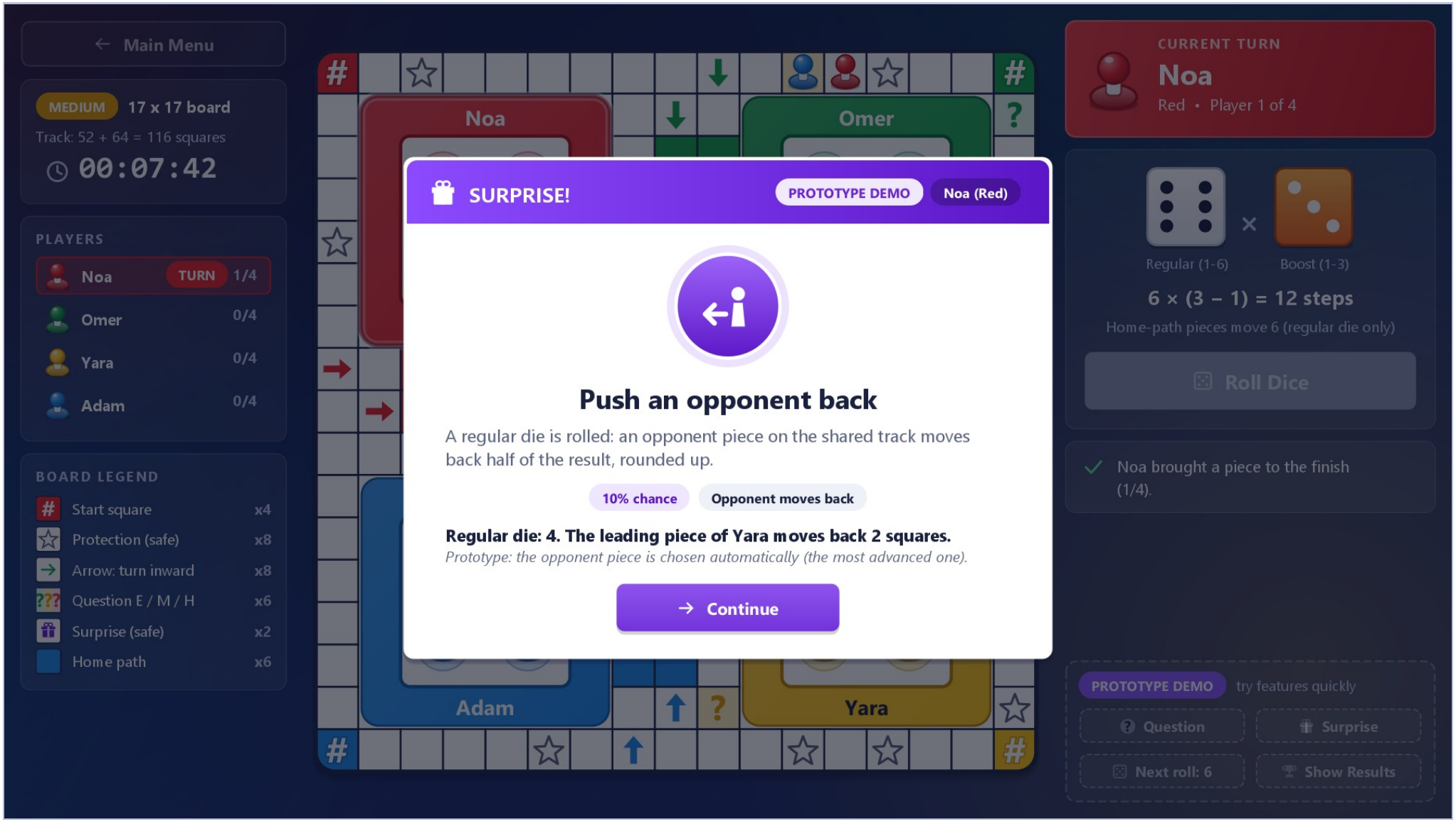
10. Answer Feedback

Feedback after Submit: the chosen answer is marked wrong, the correct answer is shown and the piece moves back by the level's steps (forward when correct). (FR-35, FR-36)



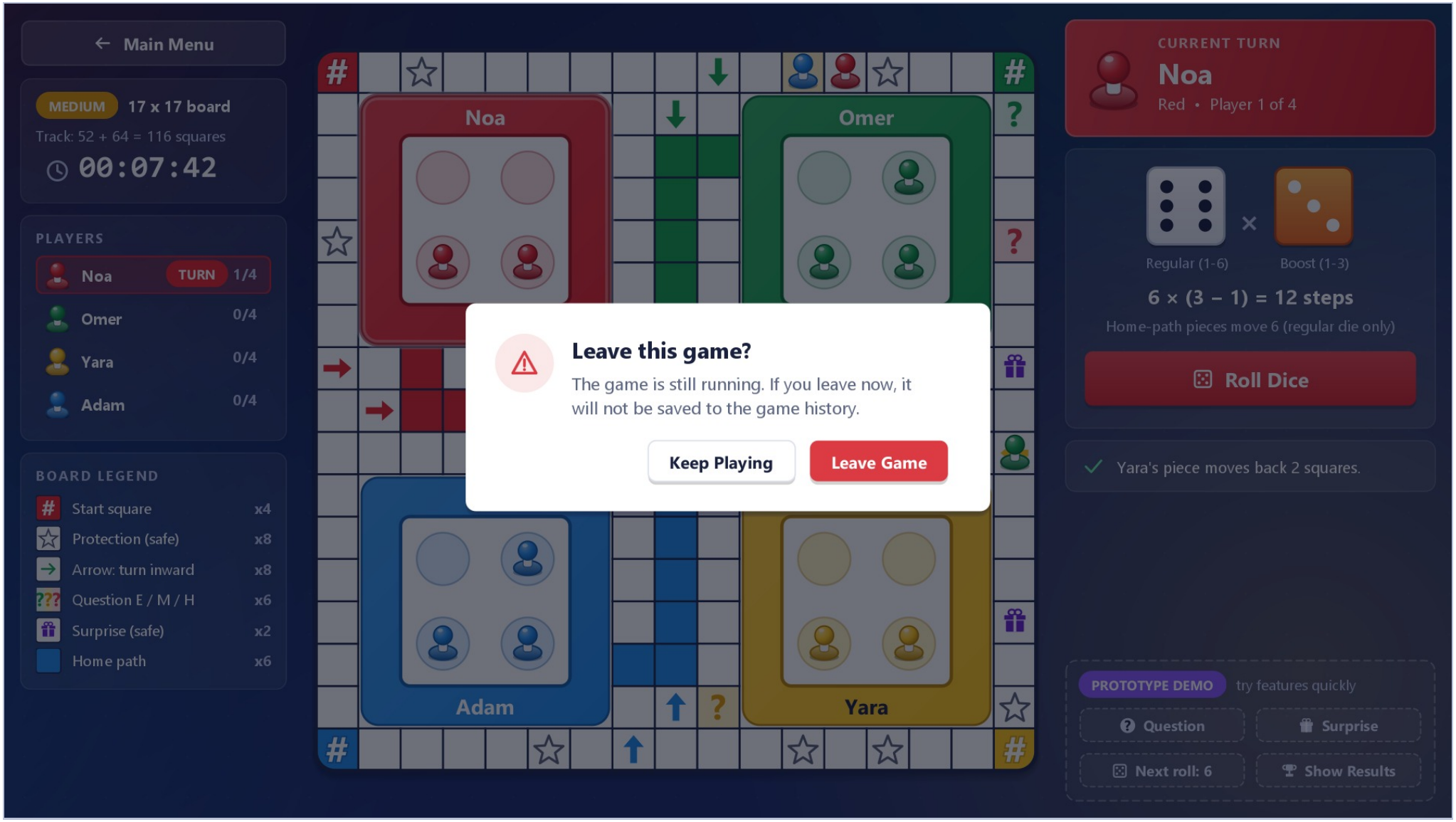
11. Surprise Event

A surprise square reveals one outcome of SRS Table 2 with its chance and effect. Behaviour that is simplified in the prototype is labelled on the card. (FR-39, FR-40)



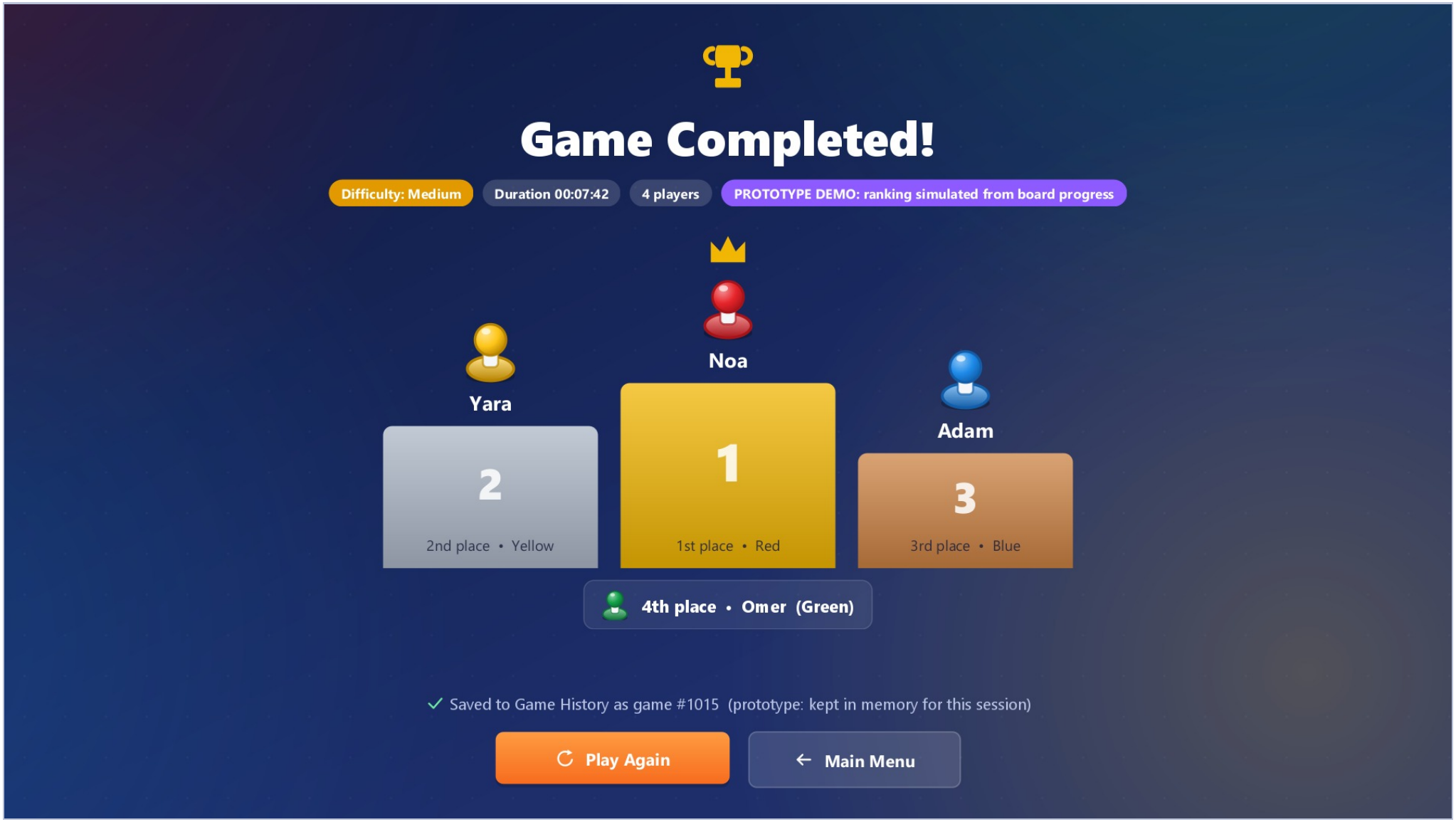
12. Leaving a Running Game

Back to Main Menu during a game asks for confirmation and explains that the game will not be saved. (FR-03)



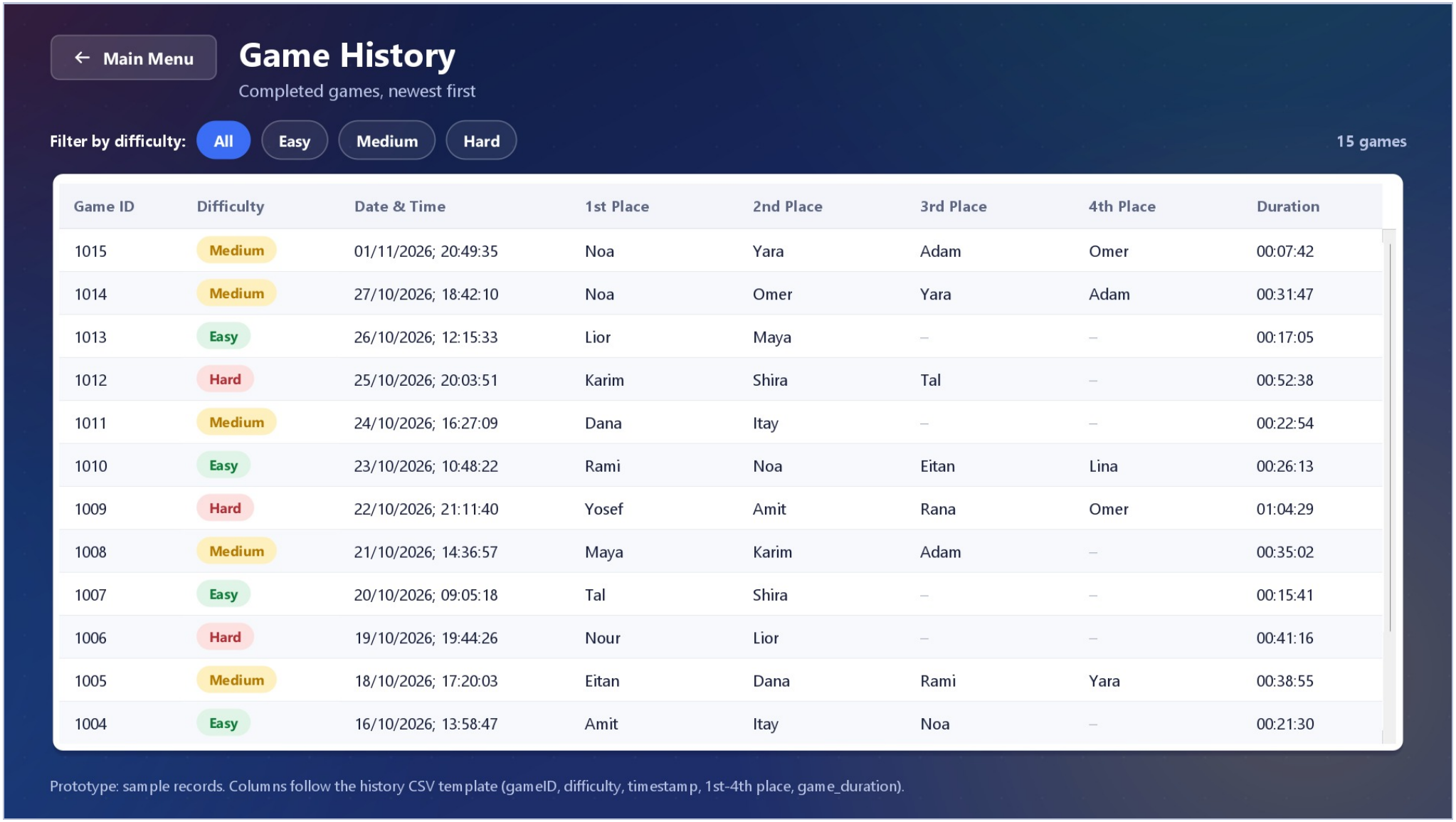
13. Game Results

Podium with rankings, player colours, difficulty and duration, with Play Again and Main Menu. In the prototype the demo panel can end a game early; the ranking is then simulated from board progress and labelled as such. (FR-45, FR-47)



14. Game History

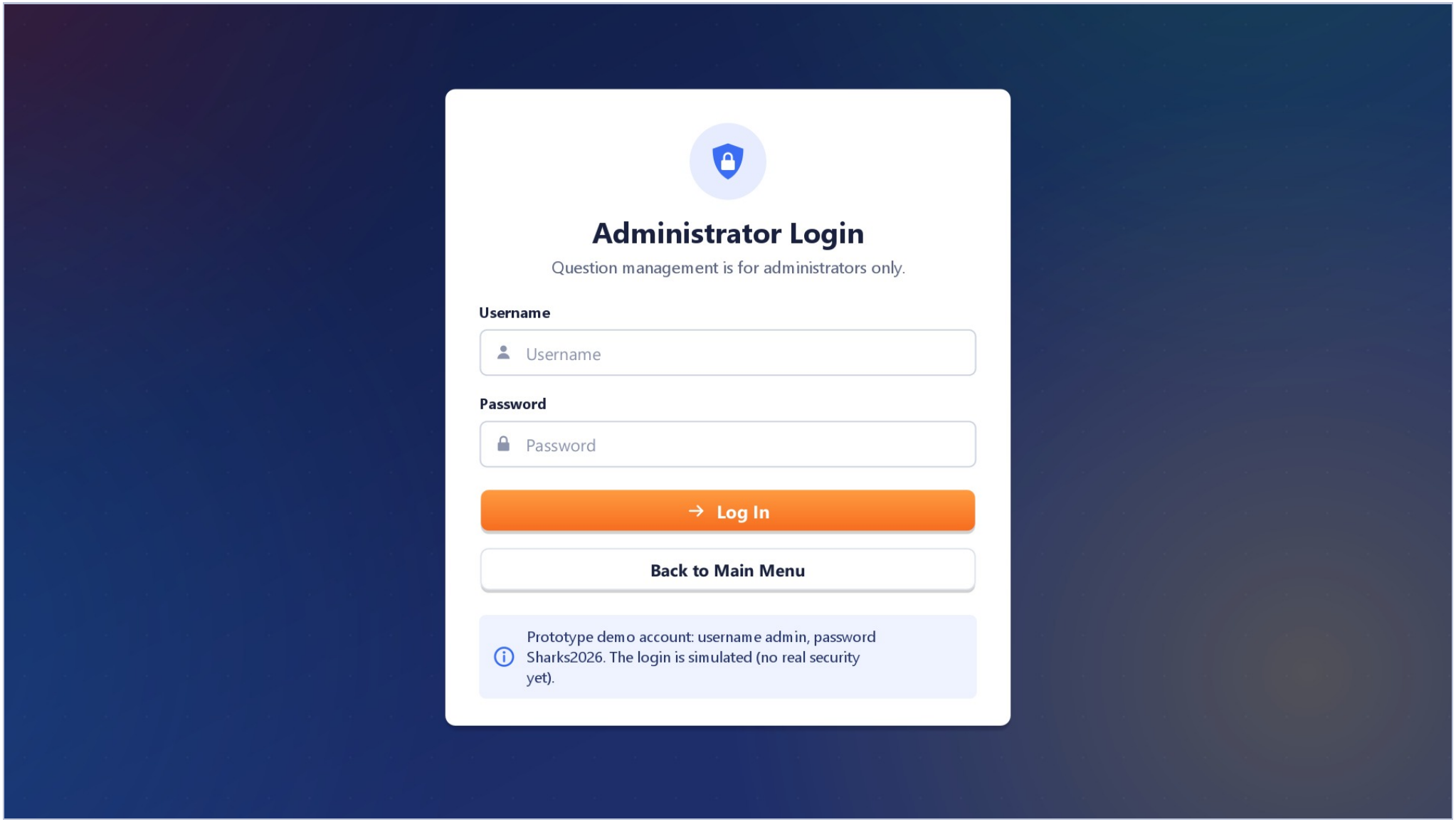
Completed games, newest first, with the fields of the history CSV template (game ID, difficulty, date and time, 1st - 4th place, duration) and a difficulty filter. The game just played is at the top. (FR-48 - FR-50, FR-63)



Game ID	Difficulty	Date & Time	1st Place	2nd Place	3rd Place	4th Place	Duration
1015	Medium	01/11/2026; 20:49:35	Noa	Yara	Adam	Omer	00:07:42
1014	Medium	27/10/2026; 18:42:10	Noa	Omer	Yara	Adam	00:31:47
1013	Easy	26/10/2026; 12:15:33	Lior	Maya	–	–	00:17:05
1012	Hard	25/10/2026; 20:03:51	Karim	Shira	Tal	–	00:52:38
1011	Medium	24/10/2026; 16:27:09	Dana	Itay	–	–	00:22:54
1010	Easy	23/10/2026; 10:48:22	Rami	Noa	Eitan	Lina	00:26:13
1009	Hard	22/10/2026; 21:11:40	Yosef	Amit	Rana	Omer	01:04:29
1008	Medium	21/10/2026; 14:36:57	Maya	Karim	Adam	–	00:35:02
1007	Easy	20/10/2026; 09:05:18	Tal	Shira	–	–	00:15:41
1006	Hard	19/10/2026; 19:44:26	Nour	Lior	–	–	00:41:16
1005	Medium	18/10/2026; 17:20:03	Eitan	Dana	Rami	Yara	00:38:55
1004	Easy	16/10/2026; 13:58:47	Amit	Itay	Noa	–	00:21:30

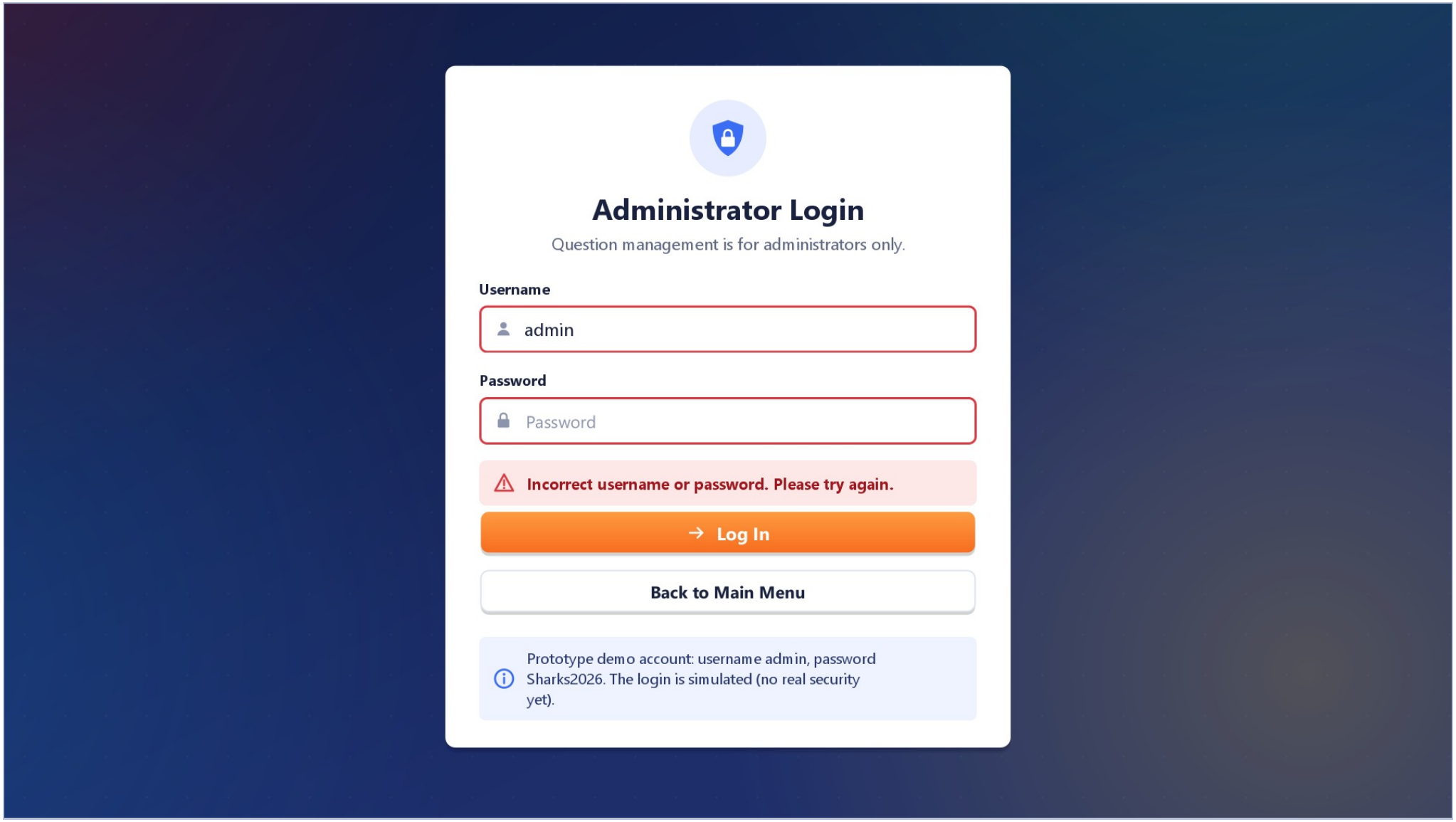
15. Administrator Login

Question Management is protected by an administrator login. The prototype uses a simulated demo account, shown on the card so that testers can continue. (FR-51, FR-52)



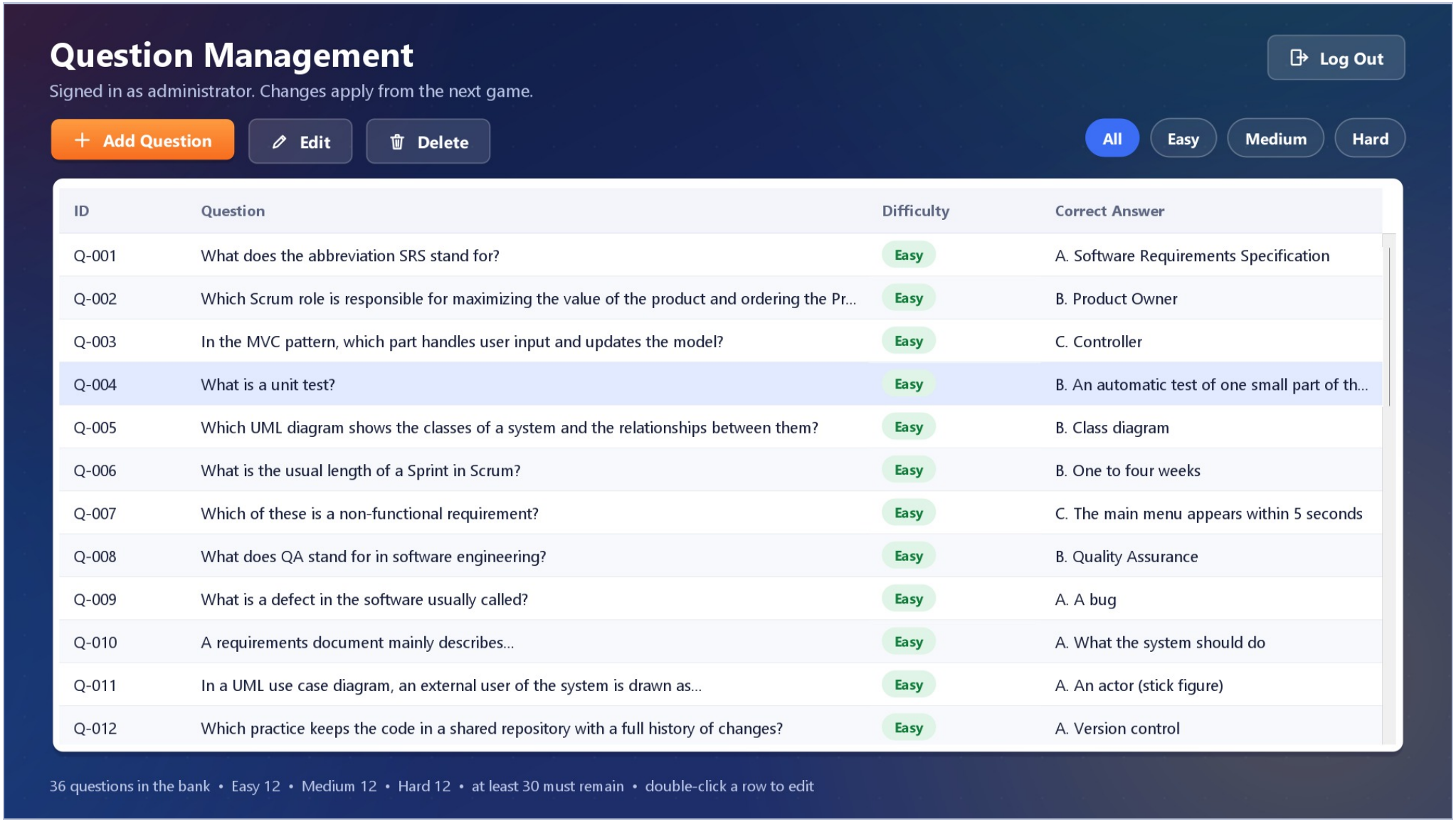
16. Login Error

Wrong credentials give a general error that does not reveal which field was wrong; three failures lock the login for 30 seconds. (FR-53, NFR-17)



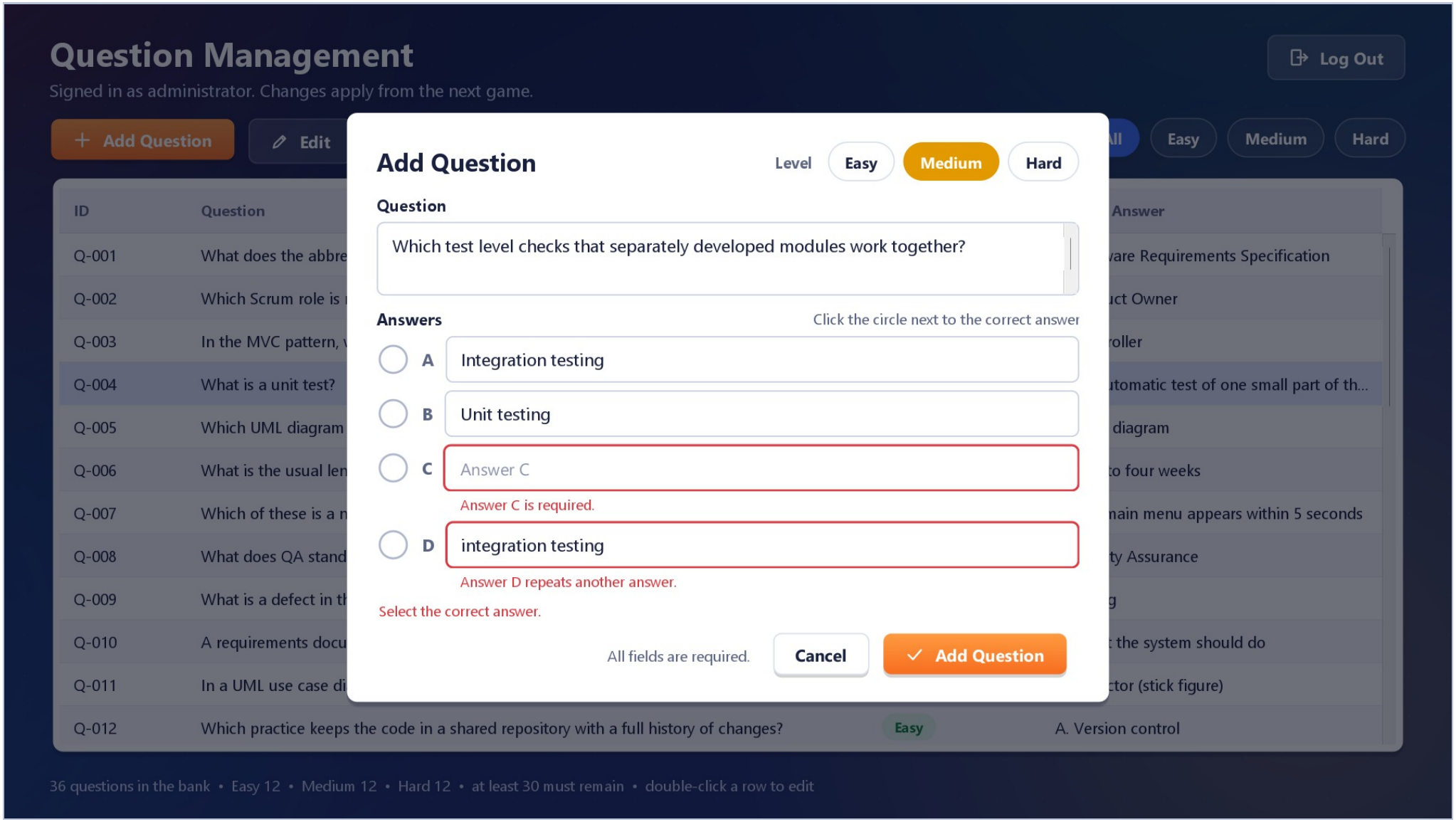
17. Question Management

All questions with ID, text, difficulty and correct answer; add, edit, delete (with confirmation, keeping at least 30 questions), filter by level and Log Out. (FR-56 - FR-62)



18. Question Form Validation

Add/Edit form: a missing answer, a duplicate answer and a missing correct answer are reported next to the fields; nothing is saved until the form is valid. (FR-58, FR-59)



Tools Used

Java 17 (Eclipse Temurin JDK)	Programming language required by the course. Compiled with javac --release 17 so it runs on any computer with Java 17 or later.
Java Swing and Java2D	Standard Java library for the desktop user interface and the custom-drawn Ludo boards, pieces, dice and icons. No external libraries, fonts or images, so the prototype works offline.
javac, jar, build.bat and run.bat	Compiling and packaging into one runnable JAR, and starting it with a double-click on Windows.
Claude Code (Anthropic AI assistant)	Used to analyse the SRS and the project description, plan the prototype, write the Java code, the self-test, the screenshot tool, this PDF and the documentation. The team reviewed and tested the result.
Visual Studio Code	Editing, reviewing and running the project.
Built-in screenshot and self-test tools	java -jar EducationalLudo.jar --screenshots creates the screenshots in this document; --selftest checks the boards, dice, rules and every user flow.
Python with PyMuPDF and Pillow	Used once to extract the official board diagrams from the project description at full resolution, to compare the drawn boards with them.
Public link	Not applicable: the prototype is an offline desktop application (EducationalLudo.jar + run.bat) that is handed in with this document.